Reliable and Invalid: Anatomy of a Gamed Verifiable Reward

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Abstract

Verifiable rewards are trusted because they are deterministic: a program either awards the points or it does not. We post-train Qwen2.5-7B-Instruct with GRPO on NYT Connections under a reward with a cheap structural component (well-formed output) and an expensive semantic component (correct groupings). On a strictly chronological test set of 162 puzzles (648 group slots), scored in one serving session, capability runs $26 \to 52 \to 80 \to 4$ slots across the untrained Instruct model, supervised fine-tuning, GRPO step 50, and the GRPO endpoint: paired stage gains of $+0.161$ and $+0.173$ on the 0–4 scale, then a fall to the 1.4-slot floor of a uniformly drawn valid partition, -0.136 $[-0.198, -0.080]$ below the untrained model, reproduced across three seeds. The *training* reward reads the destruction as mastery: it reaches its 1.6 ceiling by step 125 with zero across-rollout variance and stays there. Scoring the endpoint offline on its own training puzzles — 21 of 3,228 slots — rules out memorization and exposes the cause: a one-line bug in our GRPO data loader presented each board's sixteen words in answer-key order, so the training task, as presented, was solvable by copying the words four at a time. The collapsed policy does exactly that on 91.4% of training boards and 92.4% of test boards; a copy of a shuffled board is a random valid partition, which is why the endpoint lands on the chance floor. Entropy collapse, cross-seed convergence, and the reward ceiling are signatures of one deterministic copy rule; the structural climb is older, over half of it predating any measurable copying. The reward function was correct on every string it scored; the task presentation was gameable, and every in-sample signal read the gaming as mastery. The same reward held out, and three blind LLM judges, catch the reversal; four lines of string matching then identify the rule. Determinism certifies the scorer, not the construct: hold the reward out.

1 Introduction

Post-training against verifiable rewards is attractive for a reason that sounds decisive: the reward is a program. It returns exactly what it was written to return, and the argument that this buys immunity from reward hacking[1] rests on that reliability. This paper is about the gap reliability does not close: the gap between the reward and the construct it stands in for. We train on NYT Connections — sixteen words, partitioned into four groups of four — chosen because well-formedness and correctness are unusually separable: a response can be a perfectly valid partition and completely wrong. Our reward pays for both; two of its five components can be checked by reading the string, three require the answer key.

For fifty steps it works. In one serving session, held-out groups recovered run $26 \to 52 \to 80$ across the untrained Instruct model, the supervised checkpoint GRPO is initialized from, and step 50 — paired stage gains of $+0.161$ $[+0.056, +0.272]$ and $+0.173$ $[+0.074, +0.284]$, the reinforcement-

[1] We follow the common usage *reward hacking* throughout; Skalse et al. [2022], whose formalization we adopt, publish it under the name *reward gaming*.

learning stage contributing as much as supervised fine-tuning (§7). Over the following 353 steps the same run falls to 4 slots, below the untrained model and onto the 1.4-slot floor of a uniformly drawn valid partition, while the invalid-output rate falls by an order of magnitude and the training reward reaches its 1.6 maximum and averages 1.5814 thereafter, with zero variance across the $K = 8$ rollouts on $\sim$95% of prompts. A practitioner watching that curve saw a solved task. The curve was not lying about the reward; it was lying about the task. A control run scoring the endpoint on its own training puzzles returned 21 of 3,228 slots — not memorization — and led to the mechanism: our GRPO data loader, unlike our SFT and evaluation code, presented each board's words in answer-key order, and the converged policy simply copies the prompt four words at a time (91–92% of boards; §7). The failure is not in the reward function, which is correct; it is in what the training distribution let the reward certify, and in the practice of reading the certificate in sample.

Contributions. A fully verifiable case study in which GRPO improves and then destroys the target construct, ending below the untrained model across three seeds (§3); a complete post-hoc diagnosis: an answer-ordered prompt bug made the training task gameable, and the converged policy is a prompt-order copier on 91–92% of boards (§7); an account in which entropy collapse, the reward ceiling, and cross-seed convergence are signatures of that one rule (§5), surviving a normalization ablation (§6); and the lesson demonstrated end to end: every in-sample signal read gaming as mastery; the reward held out, and three blind judges, caught it (§7–§8).

2 Task, pipeline, and the reward's asymmetry

Task and data. 1,078 dated NYT Connections puzzles (1,077 after validation), split strictly chronologically: 807 train (2023-06-12 to 2025-08-27), 108 validation, 162 test (2025-12-15 to 2026-05-29); every test puzzle postdates every training puzzle. Date regressions of groups-correct across the test window are flat for both base (-0.158 groups/year $[-0.634, +0.292]$) and the trained policy ($+0.035$ $[-0.271, +0.329]$): contamination controlled, though the intervals are wide and only large drift would show.

Pipeline. Qwen2.5-7B-Instruct (and 1.5B for the scale contrast), rank-16 LoRA. SFT for 3 epochs on the training split, then GRPO [Shao et al., 2024] from the SFT checkpoint: $K = 8$ sampled completions per puzzle at temperature 0.9, group-relative advantages, $\beta = 0.001$ against the SFT reference, learning rate 5×10^{-6}, one epoch (403 optimizer steps) at 7B.

The leak, disclosed up front. SFT data and every evaluation in this paper present each board's sixteen words deterministically shuffled (seeded by puzzle id), precisely so group order is not a giveaway. Our GRPO loader did not: a one-line bug built training prompts from the answer key, so during reinforcement learning — only there — the words arrived grouped, positions 1–4 the first group, and so on: every GRPO prompt was solvable by a positional copy rule worth the full 1.6. Discovered post hoc (§7); disclosed here because it causes everything the training-time sections describe. Held-out measurements are unaffected: evaluation prompts were always shuffled.

The reward. Table 1 is the crux of the paper.

Table 1: The reward decomposition. Two components are verifiable from the emitted string alone; three require the answer key. The policy is trained on the sum.

Component	Value	Cost to check
Parses as four groups of four	$+0.1$	**cheap** — read the string
Correct groups	$+1.0 \times (\text{correct}/4)$	needs the answer key
All four groups correct	$+0.5$	needs the answer key
Group with exactly 3 correct words	$+0.05$	needs the answer key
Invalid output	-0.1	**cheap** — read the string

Maximum 1.6. The model can raise its score by getting better at the puzzle or by getting better at emitting well-formed output; only one is the construct. That asymmetry — cheap and expensive components, independently satisfiable — is the design property this paper studies.

3 The arc on held-out data

Table 2 reports all four arms of the reported 7B run on the 162-puzzle test set, greedy decoding, served in a single vLLM session so every comparison is within-session. The supervised checkpoint is included because it, not the untrained model, is what GRPO was initialized from.

Table 2: 7B on the test split (162 puzzles, 648 group slots; all four arms in one serving session; greedy). Means on the 0–4 scale are exact ratios of the count column. Bootstrap intervals are percentile over puzzles, $B = 1{,}000$ resamples, seed 0. This is the session §7 also scores on the training split.

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward	Solves
Instruct (untrained)	0.1605 $[0.105, 0.222]$	26	6.8%	0.165	0/162
SFT	0.3210 $[0.216, 0.432]$	52	22.2%	0.191	2/162
GRPO, step 50	**0.4938** $[0.383, 0.623]$	**80**	**16.0%**	**0.252**	2/162
GRPO, step 403	0.0247 $[0.000, 0.056]$	4	0.6%	0.125	0/162

Two reference points frame the table. A uniformly drawn valid partition recovers $4 \times 5775/2{,}627{,}625 = 0.0088$ groups per puzzle, or **1.4 slots** across this split; the step-403 endpoint sits at 2.8 times that floor — and §7 shows why: its copied responses implement that chance process. Second, because all four arms sit in one session, the stage accounting is paired: $26 \to 52 \to 80 \to 4$, the supervised stage contributing $+0.1605$ $[+0.0556, +0.2716]$ and the reinforcement-learning stage $+0.1728$ $[+0.0741, +0.2840]$ — near-equal gains, both intervals excluding zero.

The method works, then reverses. Both movements are paired within Table 2's session: the climb to step 50 is the two stage gains just given, each excluding zero, and step 403 falls below the untrained model by -0.1358 $[-0.1975, -0.0802]$. The earlier three-arm session read $+0.315$ $[+0.204, +0.438]$ and -0.136 $[-0.191, -0.080]$ for the same two contrasts; means in Table 2 are exact ratios of the count column. Two details locate the mechanism. Held-out mean reward reads 0.252 at step 50 and 0.125 at step 403 against base's 0.165 (a separate session reads step 100 at 0.2664 to a step-50 re-serve's 0.2525), while the training reward reaches its ceiling by step 125 (§5): the same reward ranks the checkpoints correctly out of sample and fails to on the training curve — that contrast, not the reward's determinism, is this paper's subject. And the step-50 policy is worse than base on structure (16.0% invalid against 6.8%) but *not* worse than its actual initialization, the SFT arm at 22.2% — so what is specific to the phase after step 50 is the semantic collapse, not the structural climb, which was already more than half complete before the peak (§5).

How wide is the peak? In a separate serving session, step 100 evaluates at 76/648 against a step-50 re-serve's 80/648, paired difference -0.0247 $[-0.1481, +0.0927]$ ($n = 162$). We read this as uninformative rather than as equivalence: the interval's width is comparable to the entire step-50 effect, so it is equally consistent with step 100 matching the peak and with step 100 having lost a third of it, and we did not pre-specify an equivalence margin. The normalization ablation of §6, this paper's only independent trajectory, argues for the second reading — step 100 reads 17/400 against step 50's 35/400 on the entropy sweep and 35/432 against 62/432 on the greedy checkpoint sweep, roughly half the peak gone in both. What both sessions agree on is narrower and is all we claim: the collapse falls between steps 100 and 150, where the validation semantic score drops 35/400 $\to$ 3/400 (§5).

Reproducibility across serving sessions. Base and the final checkpoint reproduce per-puzzle *exactly* across three vLLM sessions (0/162 differences each — a deterministic copier and a converged base leave no argmax ties for batched inference to resolve). Step 50 does not: 2/162 puzzles across sessions (77 vs. 80 of 648), 6/162 within one — mid-training policies sit near ties, which is why every step-50 number in this paper carries its serving session. Table 2's 80/648 is the four-arm session used throughout; an earlier three-arm session read 77/648, and the SFT arm decodes to 52 there and 56 in a third (Appendix C). No conclusion in this paper turns on the difference.

Robustness of the endpoint. Three independent 7B seeds land at 4, 7 and 11 recovered slots of 648 (0.045 ± 0.022 on the 0–4 scale) — all below base's 26. Their LoRA update directions agree pairwise (cosine $+0.67$ to $+0.69$, merged weight deltas) — read descriptively, three runs finding the same attractor, the copy rule (§7); seeds sharing an initialization and dataset are not isotropic draws, so no random-direction null applies. Best-of-16 sampling does not rescue the converged policy: a copier has nothing new to sample (single-seed, illustrative contrasts in Appendix C).

Conditioning on well-formed output widens the gap rather than explaining it, and we label the operation for what it is. Is the effect an artifact of validity, since invalid generations score zero groups and validity is what GRPO improved? Restricting to structurally valid generations at 7B, each arm's conditional computed within a single serving session (the SFT arm's is its 56-slot session): base **0.172** ($n = 151$ of 162), SFT **0.444** ($n = 126$), GRPO **0.025** ($n = 161$). This is a rescaling and we label it as one: each conditional is the arm's unconditional mean divided by its validity rate, so the gap widens automatically when the low-scoring arm is the more valid, as here. It licenses only the negative claim, the one we need: the deficit survives with formatting held fixed. Among answers that parse, the converged policy recovers 4 group slots where the untrained model recovers 26.

The loss shows no stratum gradient we can resolve. Relative loss from the untrained model to the endpoint runs 67–100% across all four stratum labels (values in Appendix D). We registered this, pre-diagnosis, as a failed prediction of a selective-inference story; post-diagnosis it is the copy rule's expected signature, since a positional rule is blind to stratum (§7). The comparison cannot do more than that: the strata hold 21–69 test puzzles each, the untrained model's per-stratum base rates run as low as one recovered slot, and we report no intervals — a design with no power to separate uniform loss from a gradient. Read it as consistent with stratum-blindness, not as evidence for it.

4 The scale contrast at 1.5B

At 1.5B the instrument reads nothing. The untrained model recovers one group slot of 648, GRPO one, and SFT — the best arm — two, against Section 3's 1.4-slot chance floor. No arm is distinguishable from that floor (SFT's two slots carry an exact binomial interval of $[0.0015, 0.0444]$, Appendix E, which contains it), so no claim about the construct is available at this scale. What is visible is the reward driving its verifiable component: invalid output 32.1% $\rightarrow$ 2.5%, mean reward 0.049 $\rightarrow$ 0.113 (Appendix E). We report this as an observation: the runs are epoch-asymmetric (806 steps, two epochs, against 403 and one), we have no 1.5B checkpoint trajectory, and "the construct was untouched at 1.5B" would be an endpoint claim, exactly what this paper shows cannot settle the question.

5 Trajectory of the collapse: entropy, KL, and the training signal

Figure 1 scores checkpoints of the reported run on the $n = 100$ temperature-0.9 *entropy sweep* (denominators of 400; full per-checkpoint values in Appendix B); the ablation's *checkpoint sweep* (§6) is $n = 108$, greedy, /432 — different measurements, never mixed.

The semantic peak has a location; the structural climb does not. Semantic rises 0.0325 $\rightarrow$ 0.0950 from the SFT initialization to step 50 (13/400 $\rightarrow$ 38/400 slots), falls to 0.0075 by step 150, and holds at 0.0100 from 200 on. Structural (valid-partition fraction, not shown) climbs 0.37 $\rightarrow$ 0.68 $\rightarrow$ 0.64 $\rightarrow$ 0.95–0.96 across init/50/100/150 on — one reversal, at 100 — with 0.31 of its 0.59 total gain banked by step 50, concurrent with the semantic peak.

The trajectory is not a trade with a turning point: a broadly monotone structural climb (one reversal, at 100) with a semantic hump on top, and only the hump has a location. One gap qualifies the shape: no checkpoint exists between steps 100 and 150 — 81% of the KL displacement (8.12 $\rightarrow$ 46.42 nats) — so the descent is interpolated, not measured.

Entropy collapse is the rule's signature. Policy entropy falls 13.4-fold between steps 50 and 150, the window in which the semantic score falls 12.7-fold; by step 150 the policy is effectively deterministic. With §7's diagnosis in hand the reading is direct: a positional copy rule is deterministic, blind to the board's content, and reliably well-formed: entropy collapse and semantic destruction are two views of one convergence. The structural climb is not — over half of it ($0.37 \rightarrow 0.68$ by step 50) predates measurable copying (0.3% at step 50, Appendix D): format gains the rule later locks in. What the collapse removes is within-prompt variation — nothing new to sample (the pass@16 result),

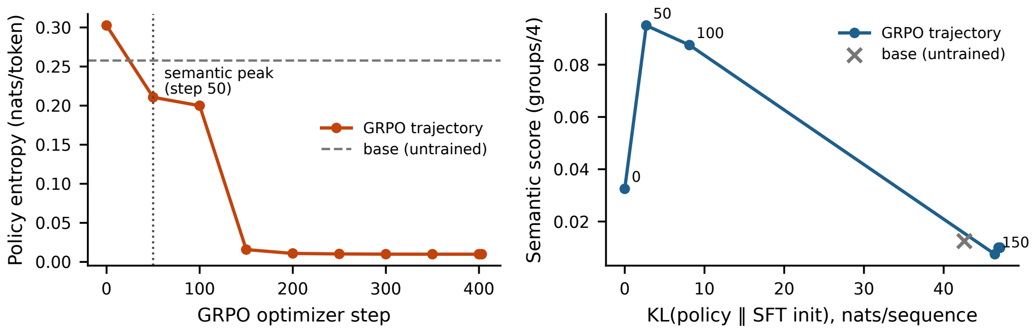

Figure 1: The mechanism, on the $n = 100$ temperature-0.9 entropy sweep (reported 7B run, seed 0). **Left:** policy entropy against optimizer step — a 30-fold collapse, with the cliff between steps 100 and 150. **Right:** semantic score against KL from the SFT initialization: the peak arrives by 2.70 of an eventual 47.05 nats (5.7%; step 50 is the earliest checkpoint retained), the remainder undoes it. The untrained model is an off-trajectory reference at 42.60 nats: KL measures displacement, not damage (§5).

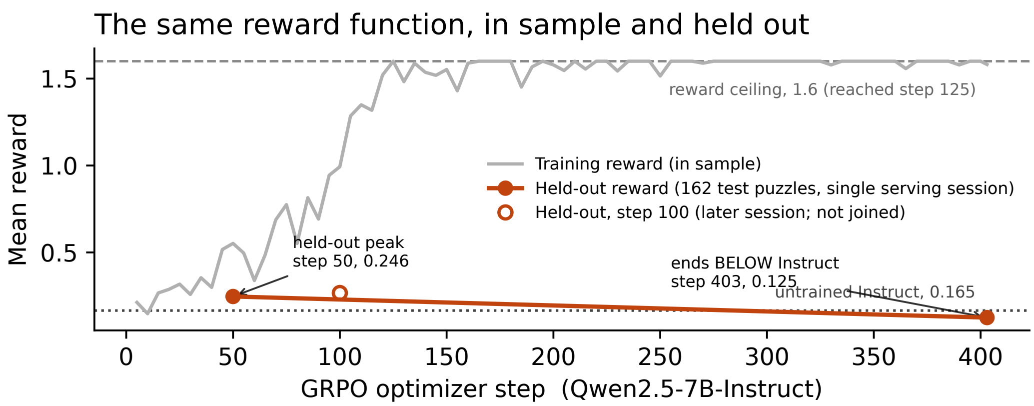

Figure 2: The same reward, in sample and held out (reported 7B run, seed 0). Training reward (gray; sampled T=0.9, K=8; TRL 5-step averages; measured on the answer-ordered prompts of §2) first touches its 1.6 ceiling at step 125, averaging 1.5814 thereafter. Held-out mean reward (162 test puzzles; solid markers; greedy; one session) peaks at step 50 in its session and ends below the untrained model; the hollow step-100 point is a later session (0.2664, vs. a step-50 re-serve at 0.2525) and is not joined.

no group-relative learning signal (the zero-advantage fraction). And what survives on held-out boards is measured twice over: the cheap structural term ($0.1(1 - 2 \cdot \text{invalid})$, Table 2) is 52% / 27% / 79% of held-out reward across base / step 50 / endpoint, and the copy rule itself is visible in 92.4% of the endpoint's test generations (§7).

KL is the budget view, and it is a path length, not a damage scale. The step-50 checkpoint sits 2.70 nats from the SFT initialization and is, jointly with step 100 (§3), the best policy this run produced on held-out data. The run spent 47.05: 80 slots at 5.7% of the displacement, 4 at all of it. Two qualifications keep this a path description, not a scale: step 50 is the earliest checkpoint retained, so "at 5.7%" means *at or before* it; and KL measures displacement, not damage — the untrained model sits 42.60 nats out on the same estimator (90.5% of the run's displacement) while scoring 0.0125 on this sweep (Appendix B), each estimate under its own policy's samples, so base is off this trajectory and 5.7% dates the peak within this run only. The sweep also compresses base's margin over the endpoint (5/400 vs. 4/400 here; 26 vs. 4 on the greedy test instrument): temperature-0.9 sampling costs a high-entropy policy far more than a collapsed one.

The training signal, read from the trainer's own logs. Figure 2 shows the paper's central contrast on one axis: the training reward first touches the reward function's ceiling of 1.6000 at step 125 and averages 1.5814 over the logged points from then on, with excursions that shrink as training proceeds: the deepest is 1.4300 at step 155, the halves average 1.5663 then 1.5960, and the trend is real by a nonparametric test (Mann–Kendall $z = 2.95$, $p = 0.003$; bounded metric, no line fit) — the copy behavior keeps sharpening long after the group-relative signal is mostly gone. What the figure cannot show is TRL's own `frac_reward_zero_std` diagnostic: the fraction of prompts whose eight sampled rewards are identical — zero group-relative advantage. It rises from 0.0 at step 50 to 1.0 at step 125 and averages 0.946 over the remainder of training, so from step 125 onward the optimizer receives no learning signal on nearly every prompt. The ceiling now has a cause: on answer-ordered prompts (§2) the copy rule earns 1.6 on every board, seen or unseen — which is why saturation begins mid-epoch, on first-visit batches, and why variance vanishes. Read as diagnostics, both in-sample signals bracket the collapse — the mean's climb to ceiling (0.9931 → 1.6000) and the variance's fall to zero both land between logged steps 100 and 125 — and neither signs it: saturation with zero spread is what mastery and degeneracy alike look like. Part of the climb is the instrument: as entropy collapses, temperature-0.9 sampling converges on greedy, so some of the rise is the sampling penalty vanishing (the sweep-vs-test gap, reversed). The remaining 278 steps of movement while $\sim$95% of prompts carry zero advantage are consistent with continued sharpening of the rule on the residual prompts that still do; the weight-space counterpart of that push we have not traced.

β **was numerically inert, and the bracket is not an experiment.** An earlier configuration at learning rate 10^{-6} and $\beta = 0.04$ froze the policy (KL ≈ 0.001 for 403 steps; greedy outputs byte-identical to SFT); the reported one, at 5×10^{-6} and $\beta = 0.001$, overshoots by roughly as much. Two qualifications, both weakening the pair as an experiment: the configurations differ in *two* variables at once (β by $40\times$, learning rate by $5\times$), so the pair bounds a region jointly and attributes nothing to either variable; and $\beta = 0.001$ was numerically inert — $\beta \cdot \mathrm{KL} = 0.0027$ reward units at step 50, 0.047 at convergence — so the KL reference never meaningfully restrained the policy, this paper has no evidence about a KL penalty large enough to matter, and the intermediate-β sweep remains unrun.

6 The ablation: the collapse is not the normalization default

We inherited `scale_rewards="group"`, verified as the trl 1.10.0 default. Reviewers of an earlier draft independently named it as a confound: near determinism, the within-group standard deviation dividing the advantage approaches zero, amplifying sampling noise into large updates — which alone might produce the collapse, its timing, and the cross-seed agreement; Liu et al. [2025] argue against the normalization on independent grounds.

We reran the 7B pipeline identically — same SFT warm start, learning rate, β, K, temperature, LoRA rank, epoch count, seed, and the same answer-ordered prompts of §2, so this ablation isolates the normalization, not the leak — with reward scaling disabled, the resolved value read back off the constructed trainer configuration. The signature is unchanged on the greedy $n = 108$ checkpoint sweep: validation semantic score rises to 62/432 at step 50, falls to 35/432 by step 100 and 4/432 by step 150, while structural validity climbs 0.70 to 0.96. The validation-selected peak (step 50, frozen before any test puzzle was scored) recovers 87/648 on test, $+0.377$ $[+0.247, +0.506]$ paired against base; the final policy 7/648, below base by -0.117 $[-0.179, -0.056]$ paired and within $+0.019$ of the original endpoint.

The signature survives in the dynamics too, though not point for point: without reward scaling, policy entropy falls 0.2111 (step 50) → 0.1346 (100) → 0.0128 (150) and stays near 0.011 through step 403 (original endpoint 0.0099), while KL rises 2.75 → 17.84 → 45.64 nats per sequence and plateaus near 46 (original 47.05; 17.84 at step 100 vs. the original's 8.12 — intermediate magnitudes differ). On the $n = 100$ temperature-0.9 entropy sweep — a different measurement from the greedy $n = 108$ checkpoint sweep above — semantic score reads 35/400 at step 50, 17/400 at 100, then 2/400 from step 150 onward. Hump, collapse, entropy endpoint, below-base endpoint: all survive with the normalization off. The phenomenon is not a library default.

Table 3: The control session: four arms, one vLLM session, greedy, both splits, shuffled evaluation prompts; percentile bootstrap, seed 0. Copy rate = emitted groups equal to consecutive quadruples of the prompt's word order (Appendix D).

Arm	Train groups (0–4)	of 3,228	Test groups (0–4)	of 648	Copy rate (test)
Instruct (untrained)	0.1871 $[0.156, 0.224]$	151	0.1605 $[0.105, 0.222]$	26	2.6%
SFT	0.5118 $[0.461, 0.568]$	413	0.3210 $[0.216, 0.432]$	52	0.3%
GRPO, step 50	0.6803 $[0.616, 0.746]$	549	0.4938 $[0.383, 0.623]$	80	0.2%
GRPO, step 403	**0.0260** $[0.012, 0.041]$	21	0.0247 $[0.000, 0.056]$	4	**92.4%**

7 The control run: not memorization — a copy rule

Training-reward saturation says the policy solves its training prompts under sampling; was that SFT-stage memorization? The control: all four arms served in one vLLM session, greedy, scored on the full 807-puzzle training split and the 162-puzzle test split under the same shuffled prompts as every other evaluation (Table 3).

Three results. The endpoint retains nothing — 21 of 3,228 training slots, indistinguishable from its own test floor: not a train/test gap, so not memorization. SFT does partially memorize, though less than the raw numbers suggest: its $+0.191$ train-minus-test gap (0.5118 vs. 0.3210 after three epochs) must be read against the untrained model's own $+0.027$ gap on the same chronological splits (0.1871 vs. 0.1605), which is split difficulty rather than memorization; the adjusted excess is $+0.164$. Step 50 inherits essentially all of it ($+0.160$ adjusted) while improving on SFT by the same margin on both splits ($+0.169$ / $+0.173$): the peak's gain generalizes. And the shared session yields the paired stage split: $26 \to 52 \to 80 \to 4$, SFT $+0.1605$ $[+0.0556, +0.2716]$, RL $+0.1728$ $[+0.0741, +0.2840]$, endpoint -0.1358 $[-0.1975, -0.0802]$ vs. the untrained model (the original session read -0.136 $[-0.191, -0.080]$).

Four lines of string matching find the mechanism. A 1.6 training reward against 21 recovered training slots forced a look at the training prompts, where the bug of §2 was found. The behavioral test counts emitted groups equal, as sets, to consecutive quadruples of the prompt's word order (definition and full counts in Appendix D). The endpoint emits such groups on 91.4% of training boards and 92.4% of test boards, with 88.2% and 90.7% of its responses pure four-quadruple copies; SFT and step 50 sit at 0.3%, below even the untrained model's weak bias (4.9% / 2.6%). The converged policy is a prompt-order copier. On training prompts, where order was the answer, copying earns 1.6; on shuffled prompts a copy is a fixed valid partition of a random order — exactly the uniformly drawn valid partition behind §3's 1.4-slot floor, and its pure-copy responses measurably are that process: 2 slots against 6.25 chance-expected on train. The aggregate sits above chance (21 vs. 7.1 expected, $p \approx 2 \times 10^{-5}$; 4 vs. 1.4 on test, same direction) — an excess carried entirely by the 4–9% of responses that are not copies: residual solving the rule has not displaced.

8 The reversal is legible from outside the reward

The 162 test generations from base, step 50 and the final policy were scored blind (board words and completion only) by three LLM judges (gpt-5-mini, gemini-2.5-flash, claude-sonnet-4-5) on a 1–10 scale, 1458/1458 responses parsed. All three rank step 50 above base above the final policy, all nine pairwise per-puzzle differences excluding zero under a paired bootstrap (seed 0; step-50 minus final: $+0.60$ $[+0.44, +0.78]$ / $+3.25$ $[+2.81, +3.67]$ / $+1.62$ $[+1.33, +1.90]$). That ordering rules out the obvious confound: the final policy is the *most* well-formed arm on this split (0.6% invalid against 6.8% and 16.0%), so a shared preference for well-formed text would have placed it first, not last. In hindsight the judges were reading the copy rule: the arm they unanimously rank last is the copier (§7). Judges do err in correlated ways [Kohli, 2026], so we read this as three correlated readings of one external signal rather than three independent confirmations; per-item Pearson r is 0.456 / 0.587 / 0.626 across the judge pairs (486 items), lowest on the final arm (0.200–0.433); the nine comparisons are one uncorrected family, all excluding zero by margins no standard correction would close.

9 Implications for evaluation practice

Distinguish in-sample from held-out proxy metrics before concluding a proxy has failed. Evaluated on held-out data, our reward ranks the three checkpoints correctly; it fails only in the form practitioners actually consult it — as a training curve, in sample, where it reads perfect. The cheap remedy is not a better reward but the same reward on data the policy was not updated against — plus the trainer's own in-sample brackets (mean saturation, zero variance), though neither signs the turn (§5). Held out, the separation is real but not large: paired against the untrained model over the 162 test puzzles, mean reward moves $+0.086$ $[+0.045, +0.129]$ at step 50 and -0.040 $[-0.061, -0.019]$ at the endpoint, both excluding zero (percentile bootstrap, $B = 1{,}000$, seed 0; SFT at $+0.026$ $[-0.015, +0.068]$ does not). We flag the margin because 79% of the endpoint's held-out reward is the cheap structural term: a reward weighted slightly differently could have kept the aggregate above base while the construct collapsed exactly as it did. The aggregate caught this run; the decomposition below is what makes it a diagnostic rather than a coincidence.

Determinism certifies the scorer, not the task. Our reward was correct on every string it scored, and the run was still gamed — through the prompt distribution. The audit cost four lines of string matching, but nothing in-sample prompted it: held-out evaluation said there was something to find. One function, two measurements: in sample, reliable and invalid — this paper's title; held out, what validity it had.

Report reward components separately, never the aggregate. This is the recommendation we would keep if we could keep only one. The failure is invisible in the sum and obvious in the decomposition, which costs nothing: §5 reports it for our three arms (52%, 27%, 79%). It is a standard reward-model diagnostic, simply not applied to deterministic rewards: determinism is taken to have settled the question.

Treat single-seed RL results as uninterpretable. The caution applies to results that come out well too, including step 50: one seed (endpoint spread across seeds: 0.025–0.068), the argmax of a nine-point validation grid (step 100 reads 0.0875 to its 0.0950 at $n = 100$, inside one standard error). The grid preceded every test evaluation of the selected checkpoint — ordinary validation-based selection. Whether step 50 is a point or a shoulder we cannot say: §3's step-100 evaluation is too wide to separate the two, and the ablation trajectory suggests decline has already begun by then.

10 Related work

Reward-model overoptimization. Gao et al. [2023] established this curve shape for *learned* reward models — gold-standard performance rising then declining with KL — where the explanation is Goodhart against an approximation [Stiennon et al., 2020]. Our reward is a program: it approximates nothing, and there was nothing in the *function* to game — the gaming ran through the prompt distribution (§2), a channel the learned-reward literature's remedy, a better reward model, does not touch. Determinism is the very feature cited to argue verifiable rewards escape this failure mode; here it guaranteed only that the shortcut was scored accurately.

RL for reasoning and entropy collapse. Verifiable-reward RL underlies recent reasoning systems [DeepSeek-AI et al., 2025, Shao et al., 2024]; Yue et al. [2025] ask whether it extends capability beyond the base model's pass@k ceiling, and here RLVR, initialized from an instruction-tuned stack, ends below *both* of its own antecedents on the construct. Cui et al. [2025] document entropy collapse as a central dynamic of RL for reasoning; here the collapse signals convergence onto a leaked shortcut, not an independent cause (§7). Liu et al. [2025] critique group-level advantage normalization; §6 shows the phenomenon does not depend on it.

Reward hacking and its taxonomy. Skalse et al. [2022] formalize the failure mode (as *reward gaming*); ours is gaming in their sense against the composed task — reward plus prompt distribution — not the reward function, which was exact. Abahana [2026] classifies RLHF failure transitions by directional changes in learned reward and judge scores; ours differs in instrument (a rule-based reward, exact) and in detection (bracketed by held-out data rather than flagged in flight). Krakovna et al. [2020] catalogue specification gaming across RL systems; the entry our run would occupy is an ordinary one, and we claim novelty for the trace rather than the phenomenon.

Shortcut learning and leakage. What our policy learned is a shortcut in the sense of Geirhos et al. [2020]: a decision rule that succeeds on the training distribution and fails off it, because a feature of the input — here, word order — carried the label. Lapuschkin et al. [2019] make the same point for supervised vision models under the name Clever Hans prediction. Neither literature is usually applied to RL post-training, where the training distribution is regenerated by a trainer rather than loaded from a file, and where a leak therefore hides in code rather than in data. That is the transfer we are making: a verifiable reward does not exempt a run from the failure mode, it only guarantees the shortcut is scored accurately.

Abruptness. Pan et al. [2022] report phase transitions in reward misspecification — capability thresholds at which behavior shifts qualitatively and true reward falls sharply — in some of the environments they study. Our collapse between steps 100 and 150 has that shape, and their result is the reason we treat the gap in our checkpoint grid as a real limitation (§11) rather than an interpolation detail: if the turn is a transition rather than a slope, its location is the measurement worth having.

Validity in AI evaluation. Measurement-theoretic critiques of ML evaluation [Jacobs and Wallach, 2021, Freiesleben and Zezulka, 2025] distinguish a measurement from the construct it operationalizes; Chehbouni et al. [2025] apply it to LLM judges; ours is a case where the measurement is maximally reliable and the validity gap does the damage anyway. Agarwal et al. [2021] motivate our statistical practice.

11 Limitations

The largest limitation: no leak-free run exists. Every training run here (the reported 7B, the three seeds, the ablation) consumed the answer-ordered prompts, so nothing separates what GRPO does to this construct from what it did to the leaked presentation of it; the one-line fix is in the repository, and the shuffled-prompt rerun is the experiment these conclusions wait on. One task, one reward design; the checkpoint trajectory is 7B seed 0 only, so the scale claim rests on endpoints at both scales and a trajectory at one, and an endpoint cannot rule out a traversed-and-returned 1.5B peak (§4). No human baseline exists: "worse than the untrained model" is established, "bad in absolute terms" is not. Three-seed evidence covers only the converged endpoint (spread 0.025–0.068, why we distrust single numbers); the peak's height and location may move across seeds, and our step-100 evaluation is too wide to establish that it is not a knife edge (§3). A single β is reported alongside one that froze the policy (no intermediate run, a learning-rate difference besides, $\beta \cdot \mathrm{KL} \leq 0.047$ throughout), so a working KL penalty remains untested. The diagnosis is post hoc (§5's readings are retrospective), the 1.5B loader shares the leak (copy rate unmeasured), and the blind judges are correlated scorers, not independent votes (§8).

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A The two prompts

The bug of §2 is one line, and it is easiest to see as the pair of strings the model actually received. Both are the user turn of the same puzzle (id 3, 2023-06-14), under the same system prompt; the answer key is `{CHEEK, EYE, MOUTH, NOSE}`, `{CHOW, GOBBLE, SCARF, WOLF}`, `{LAB, PEKE, PIT, POM}`, `{AMIGO, KING, STOOGE, TENOR}`.

GRPO training prompt, as the reported run consumed it (`train/grpo.py::build_dataset`):

```
Words: CHEEK, EYE, MOUTH, NOSE, CHOW, GOBBLE, SCARF, WOLF,
       LAB, PEKE, PIT, POM, AMIGO, KING, STOOGE, TENOR
```

SFT data and every evaluation (`data/formatting.py::shuffled_words`, a puzzle-seeded permutation):

```
Words: CHOW, POM, PIT, KING, AMIGO, NOSE, SCARF, CHEEK,
       EYE, TENOR, STOOGE, GOBBLE, MOUTH, LAB, PEKE, WOLF
```

In the first, emitting the words four at a time in the order given scores the full 1.6. In the second it scores what a uniformly drawn valid partition scores. Nothing else differs — same system prompt, same reward, same parser — and the difference is a missing call to the shuffle the rest of the codebase already used. The regression test we have since added asserts the two paths agree per puzzle, which is the property that would have caught this and the weaker "is not in answer order" assertion would not.

B Per-checkpoint values for the entropy sweep

Table 4 tabulates all eleven rows of the sweep artifact: the trajectory plotted in Figure 1 (including the step-400 checkpoint, indistinguishable from step 403 at this precision), the structural column discussed in Section 5, and the off-trajectory untrained Instruct reference. All values are from the $n = 100$ temperature-0.9 entropy sweep of the reported 7B run (seed 0); the artifact is `results-analysis/entropy-kl-7b.json` in the released repository, `https://github.com/jacksonmlukas/connections-rl`, which also holds the training and evaluation configs, the reward implementation, and the generation logs behind every table here.

Unless stated otherwise, all intervals in this paper are percentile bootstrap over *puzzles* — the unit of independent sampling — with $B = 1{,}000$ resamples at seed 0; per-slot counts are reported alongside means but never resampled independently, since a puzzle's four slots are not independent. Where a count is small enough that the bootstrap degenerates (Table 5), we substitute Clopper–Pearson intervals and say so.

Table 4: The reported 7B run on the $n = 100$ temperature-0.9 entropy sweep. Structural = fraction of outputs that are a valid partition; semantic = mean correct groups / 4. KL is the per-sequence sample-based estimate against the SFT initialization, each row estimated under its own policy's samples; the untrained Instruct row is off the GRPO trajectory (§5).

Checkpoint	Policy entropy	KL from SFT init	Structural	Semantic
Instruct (untrained)	0.2577	42.60	0.69	0.0125
SFT init	0.3025	0.00	0.37	0.0325
Step 50	0.2107	2.70	0.68	**0.0950**
Step 100	0.1999	8.12	0.64	0.0875
Step 150	0.0158	46.42	0.95	0.0075
Step 200	0.0109	46.88	0.95	0.0100
Step 250	0.0102	46.96	0.96	0.0100
Step 300	0.0099	47.05	0.96	0.0100
Step 350	0.0099	47.05	0.96	0.0100
Step 400	0.0099	47.05	0.96	0.0100
Step 403	0.0099	47.05	0.96	0.0100

C 7B endpoints across three seeds

Table 5 tabulates the three-seed endpoint session discussed in Section 3; the 4 / 7 / 11 spread is the basis of the single-seed caveats in Sections 9 and 11. The pass@16 contrasts referenced there are differences of $+0.920$ $[0.778, 1.062]$ for SFT over GRPO and -0.370 $[-0.475, -0.259]$ for GRPO against the untrained Instruct model — single-seed, read as illustrative only.

Table 5: 7B endpoints across three seeds (separate serving session from Table 2; the SFT arm decodes to 0.346/56 in the Table 2 session's configuration and 0.321/52 here, differing on 2 of 162 puzzles; the untrained and final-checkpoint anchor arms reproduce across sessions exactly).

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward
Instruct (untrained)	0.160	26	6.8% $[3.1, 11.1]$	0.165
SFT	0.321	52	22.2%	0.197
GRPO (3 seeds)	0.045 ± 0.022	4 / 7 / 11	$1.2 \pm 0.6\%$	0.132 ± 0.008

D Control-session copy rates and rewards

Table 6 gives the full copy-rule test behind Section 7: for each generation, an emitted group counts as a copy when it equals, as a set, a consecutive quadruple (positions 1–4, 5–8, 9–12, 13–16) of that prompt's word order. Train prompts parsed: 806 of 807 per arm; test: 162 of 162. Mean held-out rewards in this session: base 0.1802 (train) / 0.1654 (test), SFT 0.2490 / 0.1910, step 50 0.3113 / 0.2519, endpoint 0.1242 / 0.1250 — the endpoint's test values reproduce the published 0.125 exactly. The per-stratum relative loss cited in §3 (untrained model $\rightarrow$ endpoint, over the 158 of 162 test puzzles with a single stratum label): 92% on `category` ($n = 69$), 67% on `cultural` ($n = 28$), 86% on `tag-fillin` ($n = 40$), 100% on `wordplay` ($n = 21$, from a near-floor base of 0.048, one recovered slot).

Table 6: Copy-rule rates, control session (greedy; shuffled evaluation prompts). Group-level = share of emitted groups that are prompt-order quadruples; response-level = share of responses whose four groups all are.

Arm	Train (group)	Train (response)	Test (group)	Test (response)
Instruct (untrained)	4.9% (159/3224)	4.3%	2.6%	1.2%
SFT	0.3% (9/3224)	0.0%	0.3%	0.0%
GRPO, step 50	0.3% (11/3224)	0.0%	0.2%	0.0%
GRPO, step 403	**91.4%** (2946/3224)	**88.2%**	**92.4%**	**90.7%**

E 1.5B test-split table

Table 7 tabulates the 1.5B arms discussed in Section 4; every mean is the exact ratio of its count to 162.

Table 7: 1.5B on the test split (162 puzzles, 648 group slots). Groups-correct intervals are Clopper–Pearson exact binomial on the slot count of 648, scaled to 0–4 (a percentile bootstrap is degenerate at counts of 1–2); slots cluster within puzzles, so they are approximate. Invalid-rate intervals are percentile bootstrap over puzzles. Conditioned on structurally valid output (§3), the arms read: base 0.009 ($n = 110$ of 162), SFT 0.048 ($n = 42$), GRPO 0.006 ($n = 158$).

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward	Solves
Base	0.0062 $[0.0002, 0.0343]$	1	32.1% $[24.7, 38.9]$	0.049	0.0%
SFT	0.0123 $[0.0015, 0.0444]$	2	74.1% $[67.3, 80.2]$	-0.038	0.0%
GRPO	0.0062 $[0.0002, 0.0343]$	1	2.5% $[0.6, 5.6]$	0.113	0.0%